Midterm Formula Sheet

PSTAT 126: Linear Regression

Fall 2026

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1. The Simple Linear Regression Model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$$

Symbol	Meaning
y_i	Observed response for unit i
x_i	Predictor for unit i
β_0	Intercept
β_1	Slope
ε_i	Random error
σ^2	Error variance

Implied distribution: Under normality, $\quad Y_i \sim N(\beta_0 + \beta_1 X_i,\ \sigma^2)$

Model implications:

$$\mathbb{E}(Y_i) = \beta_0 + \beta_1 X_i \qquad \text{Var}(Y_i) = \sigma^2 \qquad \text{Cov}(Y_i, Y_j) = 0 \text{ for } i \neq j$$

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2. OLS Estimation

Key Quantities

$$S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

$$S_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}$$

OLS Estimators

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Fitted Values and Residuals

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i \qquad e_i = y_i - \hat{y}_i$$

Connection to Correlation

$$\hat{\beta}_1 = r \cdot \frac{s_y}{s_x}$$

where r is the sample correlation and s_x, s_y are sample standard deviations.

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3. Properties of OLS Residuals

$$\sum_{i=1}^{n} e_i = 0 \qquad \sum_{i=1}^{n} \hat{y}_i = \sum_{i=1}^{n} y_i \qquad \sum_{i=1}^{n} x_i e_i = 0 \qquad \sum_{i=1}^{n} \hat{y}_i e_i = 0$$

The fitted line always passes through $(\bar{x},\ \bar{y})$.

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4. Properties of the OLS Estimators

Linearity: $\hat{\beta}_1 = \sum_{i=1}^{n} a_i y_i$ where $a_i = \frac{x_i - \bar{x}}{S_{xx}}$

Unbiasedness: $\mathbb{E}(\hat{\beta}_0) = \beta_0 \qquad \mathbb{E}(\hat{\beta}_1) = \beta_1$

Variances:

$$\mathrm{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}} \qquad \mathrm{Var}(\hat{\beta}_0) = \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)$$

Gauss-Markov (BLUE): OLS has the smallest variance among all linear unbiased estimators.

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5. Estimating Error Variance

Residual Sum of Squares:

$$\mathrm{RSS} = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Mean Square Error (unbiased estimator of σ^2):

$$\hat{\sigma}^2 = \mathrm{MSE} = \frac{\mathrm{RSS}}{n-2}, \qquad \mathbb{E}(\hat{\sigma}^2) = \sigma^2$$

Residual standard deviation:

$$\hat{\sigma} = \sqrt{\frac{\mathrm{RSS}}{n-2}}$$

Standard errors:

$$\mathrm{SE}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{S_{xx}}} \qquad \mathrm{SE}(\hat{\beta}_0) = \hat{\sigma}\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}$$

MLE of σ^2 (biased): $\hat{\sigma}^2_{\mathrm{MLE}} = \dfrac{\mathrm{RSS}}{n}$

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6. Sampling Distributions

Under the normality assumption ($\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$):

$$\hat{\beta}_1 \sim N\left(\beta_1,\ \frac{\sigma^2}{S_{xx}}\right) \qquad \hat{\beta}_0 \sim N\left(\beta_0,\ \sigma^2\left(\frac{1}{n} + \frac{\bar{X}^2}{S_{xx}}\right)\right)$$

$$\frac{\mathrm{RSS}}{\sigma^2} \sim \chi^2_{n-2}$$

***t*-pivots** (replacing σ^2 with MSE):

$$\frac{\hat{\beta}_1 - \beta_1}{(\mathrm{MSE}/S_{xx})^{1/2}} \sim t_{n-2} \qquad \frac{\hat{\beta}_0 - \beta_0}{\left[\mathrm{MSE}\left(\frac{1}{n} + \frac{\bar{X}^2}{S_{xx}}\right)\right]^{1/2}} \sim t_{n-2}$$

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7. Confidence Intervals

$100(1-\alpha)\%$ **CI for β_1:**

$$\hat{\beta}_1 \ \pm\ t_{1-\alpha/2,\,n-2} \cdot \mathrm{SE}(\hat{\beta}_1)$$

$100(1-\alpha)\%$ **CI for β_0:**

$$\hat{\beta}_0 \ \pm\ t_{1-\alpha/2,\,n-2} \cdot \mathrm{SE}(\hat{\beta}_0)$$

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8. Hypothesis Tests

Test statistic for $H_0 : \beta_1 = \beta_{10}$:

$$t^* = \frac{\hat{\beta}_1 - \beta_{10}}{(\text{MSE}/S_{xx})^{1/2}} \sim t_{n-2} \text{ under } H_0$$

Test statistic for $H_0 : \beta_0 = \beta_{00}$:

$$t^* = \frac{\hat{\beta}_0 - \beta_{00}}{\text{SE}(\hat{\beta}_0)} \sim t_{n-2} \text{ under } H_0$$

Decision rules at significance level α:

Hypotheses	Reject H_0 if
$H_1 : \beta_j \neq \beta_{j0}$ (two-sided)	$\lvert t^* \rvert > t_{1-\alpha/2,\, n-2}$
$H_1 : \beta_j > \beta_{j0}$ (right-sided)	$t^* > t_{1-\alpha,\, n-2}$
$H_1 : \beta_j < \beta_{j0}$ (left-sided)	$t^* < -t_{1-\alpha,\, n-2}$

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9. Prediction Intervals

Point estimate at a new input x^*:

$$\hat{Y}^* = \hat{\beta}_0 + \hat{\beta}_1 x^*$$

Variance of fitted value:

$$\text{Var}(\hat{Y}^*) = \sigma^2 \left(\frac{1}{n} + \frac{(x^* - \bar{X})^2}{S_{xx}} \right)$$

$100(1-\alpha)\%$ **CI for mean response** $E(Y \mid X = x^*)$:

$$\hat{Y}^* \pm t_{1-\alpha/2,\, n-2} \sqrt{\text{MSE} \left(\frac{1}{n} + \frac{(x^* - \bar{X})^2}{S_{xx}} \right)}$$

$100(1-\alpha)\%$ **Prediction interval** for a new individual Y^*:

$$\hat{Y}^* \pm t_{1-\alpha/2,\, n-2} \sqrt{\text{MSE} \left(1 + \frac{1}{n} + \frac{(x^* - \bar{X})^2}{S_{xx}} \right)}$$

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10. Analysis of Variance (ANOVA)

Sums of squares decomposition: $\text{SST} = \text{SSR} + \text{SSE}$

$$\text{SST} = \sum_{i=1}^{n}(Y_i - \bar{Y})^2 \qquad \text{SSR} = \sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2 = \hat{\beta}_1^2\, S_{xx} \qquad \text{SSE} = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

ANOVA Table:

Source	df	SS	MS	$\mathbb{E}$(MS)
Model	1	SSR	MSR = SSR/1	$\sigma^2 + \beta_1^2 S_{xx}$
Error	$n-2$	SSE	MSE = SSE/($n-2$)	σ^2
Total	$n-1$	SST		

F-statistic for $H_0 : \beta_1 = 0$:

$$F^* = \frac{\text{MSR}}{\text{MSE}} = \frac{\text{SSR}/1}{\text{SSE}/(n-2)} \overset{H_0}{\sim} F(1,\ n-2)$$

Reject H_0 if $F^* > F(1-\alpha;\ 1,\ n-2)$.

Connection to the t-test: $F^* = (t^*)^2$

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11. Coefficient of Determination

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}} \ \in\ [0, 1]$$

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12. F Test for Lack of Fit

Requires replicated observations (multiple Y values at the same X).

Notation: c = number of distinct X values; n_j = replicates at group j; $\bar{Y}_j$ = group mean.

$$\text{SSPE} = \sum_{j=1}^{c}\sum_{i=1}^{n_j}(Y_{ij} - \bar{Y}_j)^2, \qquad df_{PE} = n - c$$

$$\text{SSLF} = \sum_{j=1}^{c} n_j(\bar{Y}_j - \hat{Y}_j)^2, \qquad df_{LF} = c - 2$$

$$\text{SSE}(R) = \text{SSPE} + \text{SSLF}$$

Lack of Fit F-statistic:

$$F^* = \frac{\text{MSLF}}{\text{MSPE}} = \frac{\text{SSLF}/(c-2)}{\text{SSPE}/(n-c)} \overset{H_0}{\sim} F_{c-2,\, n-c}$$

Reject H_0 (reject linearity) if $F^* > F(1-\alpha;\ c-2,\ n-c)$.

Expanded ANOVA Table:

Source	df	SS	MS
Model	1	SSR	MSR
Error	$n-2$	SSE	MSE
Lack of fit	$c-2$	SSLF	MSLF
Pure error	$n-c$	SSPE	MSPE
Total	$n-1$	SST	